

NOTICE INVITING TENDER (NIT)

SMART CITY INTEGRATED TRAFFIC MANAGEMENT SYSTEM

Tender Reference No.: SCMC/ITMS/2025-26/007

Date of Issue: 15 January 2026

SMART CITY MISSION CORPORATION

Infrastructure Development Wing

5th Floor, Municipal Bhavan, MG Road, Bengaluru – 560001

Tel: +91-80-22221234 | Email: tenders@smartcitycorp.gov.in

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SECTION 1: GENERAL INFORMATION

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1.1 NAME OF WORK

Supply, Installation, Testing, Commissioning, and 5-Year Operation & Maintenance of Integrated Traffic Management System (ITMS) including Adaptive Traffic Signal Control, AI-based Video Analytics, Incident Detection System, and Central Command & Control Centre for Bengaluru Smart City Zone-3.

1.2 ESTIMATED PROJECT COST

Rs. 42,50,00,000/- (Rupees Forty-Two Crore Fifty Lakhs Only)

1.3 EARNEST MONEY DEPOSIT (EMD)

Rs. 85,00,000/- (Rupees Eighty-Five Lakhs Only)

EMD shall be submitted as a Bank Guarantee or Demand Draft drawn in favour of

Smart City Mission Corporation, Bengaluru, valid for 180 days from bid opening date.

1.4 PERFORMANCE SECURITY

5% of Contract Value (Bank Guarantee, valid for contract period + 60 days)

1.5 DEFECT LIABILITY PERIOD

24 months from the date of Taking Over Certificate (TOC)

1.6 TENDER DOCUMENT FEE

Rs. 25,000/- (Non-refundable) via RTGS/NEFT to A/C No. 1234567890, IFSC: SBIN0001234

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SECTION 2: SCHEDULE OF KEY DATES

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- 2.1 DATE OF PUBLICATION : 15 January 2026
- 2.2 PRE-BID CONFERENCE : 25 January 2026 at 11:00 AM
- 2.3 LAST DATE FOR SUBMISSION OF QUERIES : 28 January 2026
- 2.4 RESPONSE TO QUERIES : 02 February 2026
- 2.5 LAST DATE FOR BID SUBMISSION : 20 February 2026 up to 5:00 PM
- 2.6 BID OPENING DATE (TECHNICAL) : 21 February 2026 at 11:00 AM
- 2.7 BID OPENING DATE (FINANCIAL) : To be notified after Technical Evaluation
- 2.8 WORK COMPLETION PERIOD : 12 months from date of work order
- 2.9 OPERATION & MAINTENANCE PERIOD : 5 years post-commissioning

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SECTION 3: ELIGIBILITY CRITERIA / QUALIFICATION REQUIREMENTS

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3.1 LEGAL ENTITY

The bidder shall be a company/firm registered under Companies Act 2013 or Partnership Firm or LLP with valid GST Registration and PAN.

3.2 FINANCIAL ELIGIBILITY

3.2.1 Average Annual Turnover: Minimum Rs. 30 Crores per annum for the last 3 consecutive financial years (FY 2022-23, 2023-24, 2024-25).

3.2.2 Net Worth: Positive net worth as per latest audited balance sheet.

3.2.3 Working Capital / Credit Facility: Minimum Rs. 8 Crores (solvency certificate from bank required).

3.3 TECHNICAL ELIGIBILITY – SIMILAR WORK EXPERIENCE

The bidder must have successfully completed at least ONE similar work of value not less than Rs. 17 Crore OR TWO similar works each not less than Rs. 10 Crore during the last 7 years.

"Similar Work" means: Supply, Installation, and Commissioning of Intelligent Traffic Management System / Adaptive Traffic Control System / Video Surveillance System for any Government / Smart City / Municipal Authority in India.

3.4 TECHNICAL ELIGIBILITY – OEM AUTHORISATION

Bidder shall provide OEM authorisation letters for:

(a) Adaptive Traffic Signal Controllers

- (b) AI-based Video Analytics Cameras
- (c) Variable Message Signs (VMS)
- (d) Central Control Software (SCATS/SCOOT or equivalent)

3.5 KEY PERSONNEL REQUIREMENTS

- 3.5.1 Project Manager: B.E./B.Tech (Electronics/IT) with minimum 10 years experience in traffic management or ITS projects.
- 3.5.2 Site Engineer (Traffic): B.E./B.Tech Civil/Electronics, 5 years experience.
- 3.5.3 Software Lead: B.E./B.Tech (IT/CS), 7 years experience in SCADA/ITS software development.
- 3.5.4 Network & Cybersecurity Expert: Certified (CCNP/CISSP), 5 years experience.

3.6 CONSORTIUM / JV

A consortium of maximum 2 firms is permitted. Lead member shall hold minimum 51% stake. Criteria shall apply collectively for turnover, net worth but individually for similar work experience (lead member must meet similar work).

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SECTION 4: SCOPE OF WORK

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4.1 SUPPLY AND INSTALLATION

- 4.1.1 Adaptive Traffic Signal Controllers (ATSC) at 52 intersections
- 4.1.2 IP-based PTZ cameras (2 per intersection) – 104 units
- 4.1.3 AI Video Analytics Server with edge-computing modules
- 4.1.4 Vehicle Detection Sensors (Loop/Radar) – 208 lanes

- 4.1.5 Variable Message Signs (LED, 3m x 1.2m) – 18 locations
- 4.1.6 Optical Fibre Cable (OFC) backbone – 85 km ring topology
- 4.1.7 Central Command & Control Centre (CCC) setup – servers, workstations,
video wall (4x3 LED, 55" tiles), UPS, networking equipment
- 4.1.8 Emergency Vehicle Preemption (EVP) system at 52 intersections
- 4.1.9 Pedestrian countdown signals at 52 intersections

4.2 SOFTWARE DEVELOPMENT AND INTEGRATION

- 4.2.1 Adaptive Signal Control Software (SCATS or equivalent licensed)
- 4.2.2 AI-based Incident Detection and Red Light Violation Detection (RLVD)
- 4.2.3 Automatic Number Plate Recognition (ANPR) – 104 camera feeds
- 4.2.4 Traffic Data Management System with historical analytics
- 4.2.5 Mobile app for field officers and real-time alerts
- 4.2.6 Integration with existing City Operation Centre (COC) via API

4.3 CIVIL WORKS

- 4.3.1 Foundation and pole erection at all 52 intersections
- 4.3.2 Underground conduit laying for OFC and power cables
- 4.3.3 Junction boxes, earthing, electrical connections
- 4.3.4 CCC fit-out: false ceiling, flooring, air-conditioning, fire suppression

4.4 TESTING AND COMMISSIONING

- 4.4.1 Factory Acceptance Test (FAT) in presence of client team
- 4.4.2 Site Acceptance Test (SAT) and integrated system testing
- 4.4.3 Parallel operation with existing system (minimum 30 days)
- 4.4.4 Training: 5-day structured training for 20 operator staff

4.5 OPERATION AND MAINTENANCE (5 YEARS)

- 4.5.1 24x7 NOC staffing with dedicated helpdesk
- 4.5.2 System uptime SLA: minimum 98.5% (measured monthly)
- 4.5.3 Mean Time to Repair (MTTR): 4 hours for critical, 24 hours for others
- 4.5.4 Annual preventive maintenance (4 cycles per year)
- 4.5.5 Spare parts inventory maintenance (critical spares on-site)
- 4.5.6 Annual performance reports and traffic flow analysis

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SECTION 5: BILL OF QUANTITIES (BOQ)

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SCHEDULE A – SUPPLY AND INSTALLATION

S.No	Description	Unit	Qty	Rate(Rs.)	Amount(Rs.)
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A.1	Adaptive Traffic Signal Controller (Type-I, 8-phase)	No.	52	180000	9360000
A.2	IP PTZ Camera 4MP, 30x Optical Zoom, IR 100m	No.	104	65000	6760000
A.3	AI Video Analytics Edge Server (GPU-enabled)	No.	10	450000	4500000
A.4	Vehicle Inductive Loop Detector (dual loop set)	Set	208	12000	2496000
A.5	Radar-based Multi-lane Vehicle Detector	No.	52	55000	2860000
A.6	Variable Message Sign LED (3m x 1.2m, full colour)	No.	18	350000	6300000
A.7	OFC Cable (24 Core, Armoured, Single Mode)	m	85000	120	10200000
A.8	OFC Splicing & Termination (per splice point)	No.	420	2500	1050000
A.9	Emergency Vehicle Preemption System (per jn)	No.	52	25000	1300000
A.10	Pedestrian Countdown Signal Unit	No.	208	8500	1768000

SCHEDULE B – COMMAND & CONTROL CENTRE

B.1	Server Rack (42U, with PDU, cable mgmt)	No.	8	85000	680000
B.2	Application Server (16-core, 128GB RAM, 10TB)	No.	4	650000	2600000
B.3	Storage Server (NAS, 200TB usable)	No.	2	850000	1700000
B.4	Video Wall Controller (4x3 LED, 55" tiles)	Set	1	2800000	2800000
B.5	Operator Workstation with dual monitor	No.	12	95000	1140000
B.6	UPS (60 KVA, 30-min backup)	No.	2	420000	840000
B.7	Air Conditioning (5-Ton Precision, N+1)	No.	3	280000	840000
B.8	Network Switch (48-port managed, L3)	No.	15	75000	1125000

SCHEDULE C – CIVIL AND INSTALLATION

C.1	Signal Pole (6m GI, hot-dip galvanised, with arm)	No.	208	22000	4576000
C.2	Underground conduit (100mm HDPE) with cable pull	Set	85000	85	7225000
C.3	Junction Box (weatherproof IP66, 600x400x250)	No.	312	3500	1092000
C.4	Earthing (plate earthing per IS 3043)	No.	52	4500	234000
C.5	CCC Civil Fit-out (false ceiling, flooring, partition)	Sqm	450	8500	3825000

SCHEDULE D – SOFTWARE AND INTEGRATION

D.1	Adaptive Signal Control Software (SCATS/equivalent)	LS	1	3500000	3500000
D.2	AI Incident Detection + RLVD Module	LS	1	2200000	2200000
D.3	ANPR Software with Database (104 feeds)	LS	1	1800000	1800000
D.4	Traffic Data Management & Analytics Platform	LS	1	1500000	1500000
D.5	Mobile Application (iOS + Android)	LS	1	850000	850000
D.6	COC Integration APIs and middleware	LS	1	600000	600000
D.7	Cybersecurity Hardening and Compliance	LS	1	400000	400000

D.8 Training and Documentation	LS	1	250000	250000
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SCHEDULE E – OPERATION AND MAINTENANCE (5 YEARS)

E.1 Annual Maintenance Year 1 (post-DLP)	LS	1	3200000	3200000
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E.2	Annual Maintenance Year 2	LS	1	3360000	3360000
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E.3 Annual Maintenance Year 3	LS	1	3528000	3528000
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E.4 Annual Maintenance Year 4	LS	1	3704000	3704000
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E.5	Annual Maintenance Year 5	LS	1	3889000	3889000
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TOTAL ESTIMATED COST (A+B+C+D+E): Rs. 42,50,00,000/-

NOTE: Rates are to be quoted inclusive of all taxes, GST, transportation, insurance, and incidental charges. GST is payable extra at applicable rates.

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SECTION 6: COMMERCIAL TERMS AND CONDITIONS

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6.1 PAYMENT TERMS

Stage 1: 10% on signing of Contract Agreement

Stage 2: 60% on pro-rata basis against supply and delivery (against submission of delivery challans, inspection reports, and receipt at site)

Stage 3: 20% on completion of installation, testing, and commissioning
(against issue of Provisional Acceptance Certificate – PAC)

Stage 4: 5% on issuance of Final Acceptance Certificate (FAC) / TOC

Stage 5: Remaining 5% released upon expiry of Defect Liability Period (DLP),
subject to satisfactory performance. Retained as Retention Money.

O&M Payment: Quarterly in arrears, subject to SLA compliance verification and
submission of NOC from O&M team.

6.2 LIQUIDATED DAMAGES (LD)

6.2.1 Delay in Completion: 0.5% of contract value per week of delay, subject to
maximum cap of 10% of contract value.

6.2.2 SLA Breach (O&M Phase): Rs. 25,000/- per percentage point below 98.5%
uptime in any month, subject to monthly maximum of Rs. 2,00,000/-.

6.2.3 MTTR Breach: Rs. 5,000/- per incident per hour beyond agreed MTTR for
critical components.

6.3 PERFORMANCE GUARANTEE

Bank Guarantee of 5% of contract value, valid for contract period plus 60 days.
To be submitted within 15 days of receipt of Letter of Acceptance (LOA).

6.4 PRICE VARIATION

No price variation is applicable for the supply and installation phase.

For O&M phase: Escalation of 5% per annum on O&M charges effective from
Year 2 onwards (pre-built into BOQ Schedule E).

6.5 INSURANCE

Contractor's All Risk (CAR) policy for full contract value.

Workmen Compensation Insurance as per applicable laws.

Third Party Liability Insurance: Minimum Rs. 2 Crore per occurrence.

6.6 PENALTY FOR POOR PERFORMANCE

If system uptime falls below 95% for 3 consecutive months, the Corporation reserves the right to invoke Performance Bank Guarantee and/or terminate the O&M contract after issuing 30-day cure notice.

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SECTION 7: RISK ASSESSMENT (IDENTIFIED RISKS)

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7.1 HIGH RISK ITEMS

7.1.1 Utility Conflict Risk: Existing underground utilities (water, sewage, HT cable) may obstruct OFC conduit laying. Bidder shall conduct pre-survey; Corporation will provide utility maps but accuracy is not guaranteed.
Risk: Extra cost and time for deviation of utilities lies with Contractor.

7.1.2 Night Restriction Risk: Traffic management / police permission required for night work at intersections. Delay in permissions may affect timeline.
Risk: Contractor to factor in delays; no EOT guaranteed for regulatory delays.

7.1.3 Software IP Risk: Adaptive signal control software must be licensed.
Bidder to ensure license covers perpetual use; sub-licensing by Corporation post-contract must be permitted.

7.1.4 Cybersecurity Risk: The ITMS system is classified as Critical Information Infrastructure. Vendor must comply with CERT-In guidelines and submit quarterly security audit reports.

7.2 MEDIUM RISK ITEMS

7.2.1 Supply Chain Risk: Global semiconductor shortage may affect delivery of AI-edge servers and signal controllers. Bidder to furnish confirmed supply timelines and contingency plan.

7.2.2 Integration Risk: COC integration depends on availability of legacy system APIs from the existing vendor. Bidder shall perform due diligence; delays due to legacy system limitations shall not qualify for EOT.

7.3 LOW RISK ITEMS

7.3.1 Labour Availability: Unskilled and semi-skilled labour availability may vary. Contractor to plan accommodation and logistics accordingly.

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SECTION 8: EVALUATION CRITERIA

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8.1 TWO-BID SYSTEM (Technical + Financial)

8.2 TECHNICAL EVALUATION (Pass/Fail + Technical Score)

Criterion	Max Marks

Similar Work Experience	25
Financial Capacity	15
Key Personnel CVs	20

Methodology & Work Program	20
OEM Partnerships & Certifications	10
Local Service Centre Availability	10
TOTAL TECHNICAL SCORE	100

Minimum Qualifying Technical Score: 70 out of 100

8.3 FINANCIAL EVALUATION

Lowest Valid Financial Bid (L1) wins.

Quality and Cost-Based Selection (QCBS) is NOT applicable.

Only technically qualified bidders (score ≥ 70) are eligible for financial bid opening.

8.4 NEGOTIATION

The Corporation reserves the right to negotiate with L1 bidder.

In case L1 fails to negotiate or withdraws, L2 bidder may be invited.

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SECTION 9: GENERAL CONDITIONS OF CONTRACT (GCC)

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9.1 GOVERNING LAW

This contract shall be governed by the laws of India. Any disputes shall be subject to the jurisdiction of courts at Bengaluru.

9.2 DISPUTE RESOLUTION

Disputes shall be resolved by mutual negotiation (30-day period), failing which through Arbitration as per the Arbitration and Conciliation Act, 1996.

Venue: Bengaluru. Language: English. Three-arbitrator panel.

9.3 FORCE MAJEURE

Events including natural disasters, acts of war, pandemics, or government directives beyond reasonable control of either party shall constitute Force Majeure.

Notice to be given within 7 days of occurrence.

9.4 TERMINATION FOR CONVENIENCE

Corporation may terminate contract on 60-day written notice. Contractor is entitled to payment for work satisfactorily completed prior to termination.

9.5 TERMINATION FOR CAUSE

Corporation may terminate immediately if:

- (a) Contractor becomes insolvent
- (b) Work is abandoned
- (c) Persistent SLA breach beyond cure notice period
- (d) Corrupt or fraudulent practice is established

9.6 SUBCONTRACTING

Maximum 30% of work value may be subcontracted. Prior written approval required.

Prime contractor remains fully responsible for quality and performance of subcontracted work.

9.7 CHANGE ORDERS

Corporation may order additions or deletions up to 15% of contract value.

Changes beyond 15% shall be by mutual agreement. No time or cost adjustments

without written change order.

9.8 INTELLECTUAL PROPERTY

All custom software, documentation, and configurations developed specifically for this project shall become the intellectual property of the Corporation upon final payment. Commercial off-the-shelf (COTS) software retains vendor IP.

9.9 CONFIDENTIALITY

Contractor shall not disclose any project data, traffic data, surveillance footage, or system architecture to any third party without written consent. Obligation survives contract termination for 5 years.

9.10 ANTI-CORRUPTION

Contractor warrants compliance with Prevention of Corruption Act, 1988. Any corrupt practice shall result in immediate contract termination and forfeiture of performance guarantee.

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SECTION 10: DOCUMENTS REQUIRED WITH BID

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ENVELOPE 1 – EMD AND FEES

- (a) EMD Instrument (Bank Guarantee / Demand Draft)
- (b) Tender Document Fee Receipt

ENVELOPE 2 – TECHNICAL BID

- (a) Form of Tender (duly signed)
- (b) Certificate of Incorporation / Partnership Deed / LLP Agreement
- (c) GST Registration Certificate and PAN
- (d) Audited Financial Statements (last 3 years)
- (e) Banker's Solvency Certificate (not older than 3 months)
- (f) Completion Certificates for Similar Works (duly notarised)
- (g) OEM Authorisation Letters (for all components listed in 3.4)
- (h) CVs and undertakings for Key Personnel
- (i) Technical Methodology and Work Program (PERT/Gantt chart)
- (j) List of local service centres with addresses
- (k) Self-declaration of non-debarment / blacklisting
- (l) Power of Attorney for authorised signatory

ENVELOPE 3 – FINANCIAL BID

- (a) Completed Price Schedule (BOQ Schedules A through E)
- (b) Covering Letter with total bid price

All envelopes to be sealed separately, then placed in an outer envelope,
duly superscribed with Tender Reference, Work Name, and Date of Opening.

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END OF TENDER DOCUMENT

Smart City Mission Corporation | Bengaluru

Tender Ref: SCMC/ITMS/2025-26/007

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